ELEC5305 Project Feedback One

Comparative Evaluation of STFT-Based Spectral Subtraction and Wiener Filtering for Speech Enhancement

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Brief Project Description

Environmental noise from traffic, cafes, ventilation systems, and crowds can reduce the quality and intelligibility of recorded speech. This project will investigate a practical single-channel speech-enhancement problem by implementing and comparing two classical methods: **spectral subtraction** and **Wiener filtering**. Both methods operate in the short-time Fourier transform (STFT) domain and provide transparent, computationally efficient alternatives to modern deep-learning systems.

Clean VoiceBank speech samples will be combined with stationary white noise and non-stationary cafe and traffic noise from the DEMAND database. The mixtures will be created at controlled input signal-to-noise ratios of 0 dB, 5 dB, and 10 dB. Each signal will be analysed using overlapping Hann-windowed frames. Spectral subtraction will remove an estimated noise magnitude spectrum, while Wiener filtering will apply a frequency-dependent gain derived from speech and noise power estimates. The enhanced signals will then be reconstructed using the inverse STFT and overlap-add.

The two algorithms will be evaluated using output SNR, SNR improvement, Short-Time Objective Intelligibility (STOI), spectrograms, and listening examples. The comparison will examine how noise type and input SNR affect noise attenuation, intelligibility, and speech distortion. Expected outcomes include a reproducible Python implementation, enhanced audio examples, metric tables, and visual comparisons. The project directly applies ELEC5305 concepts including framing, windowing, STFT analysis, spectral estimation, filtering, phase reuse, ISTFT, and overlap-add reconstruction while remaining feasible within the available project period.

GitHub Project Site

https://sliu2040977-hash.github.io/elec5305-project-540569223/?v=20260828